

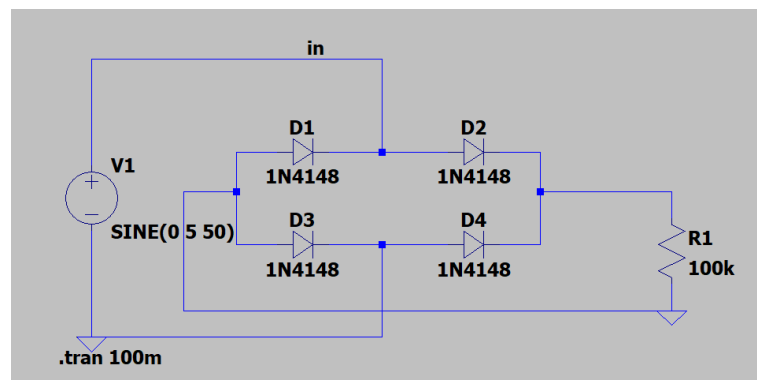
Diode Rectifier

ENGS152 Circuits Lab

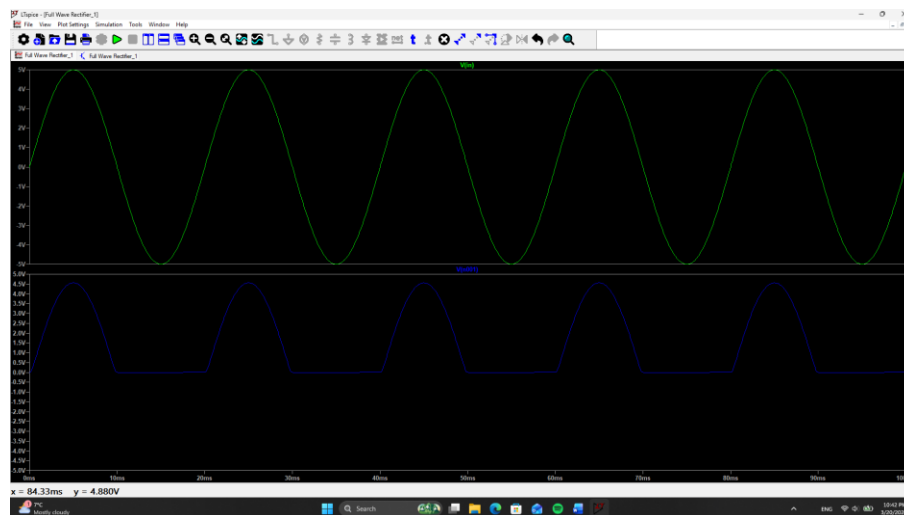
20 March 2026

Step 1: Full-wave rectifier

First, we are asked to build a full-wave rectifier circuit. You can see the circuit below or in the ltspice simulation file below. In the circuit the voltage supply must correspond to a sine voltage supply where the voltage is the length of the name (in my case Ashot, **5** letters), and the frequency must correspond to name time surname which is **50** for my name and surname. **1N4148** Diodes are used, and a resistor is added at the output to simulate a load.



First, we are asked to set the resistor value to 100 k Ω . You can see the input sine wave and the resulting output plots below.

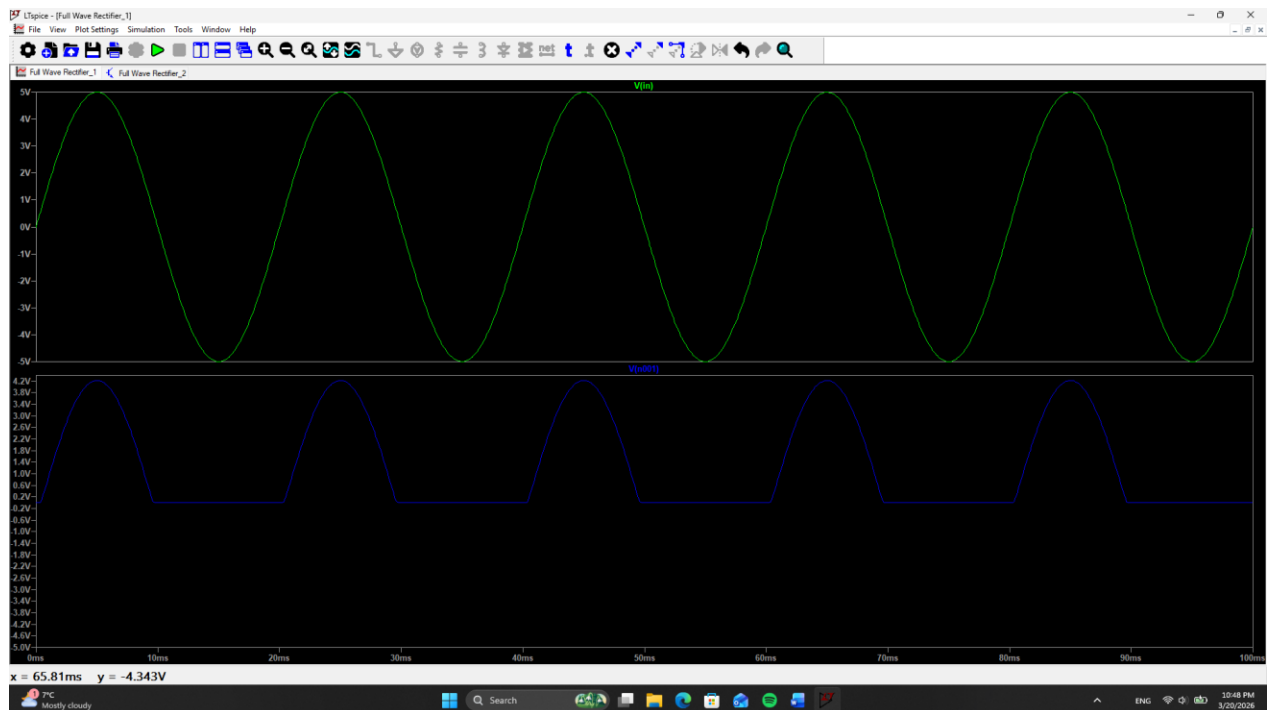


$$V_{drop} = 5V - 4.5V = 0.5V$$

$$V_{drop \text{ per diode}} = \frac{0.5}{2} = 0.25V$$

The output is full wave rectified. All negative phase of the voltage does not pass the circuit only the positive is visible at the output. The peak voltage within the circuit is changed from 5V input voltage to 4.5V output voltage.

Next, we are asked to replace the 100 kΩ resistor with 100 Ω one.



Here we can see that the output is also rectified, however the peak voltage for the output is now 4.2V.

$$V_{drop} = 5V - 4.2V = 0.8V$$

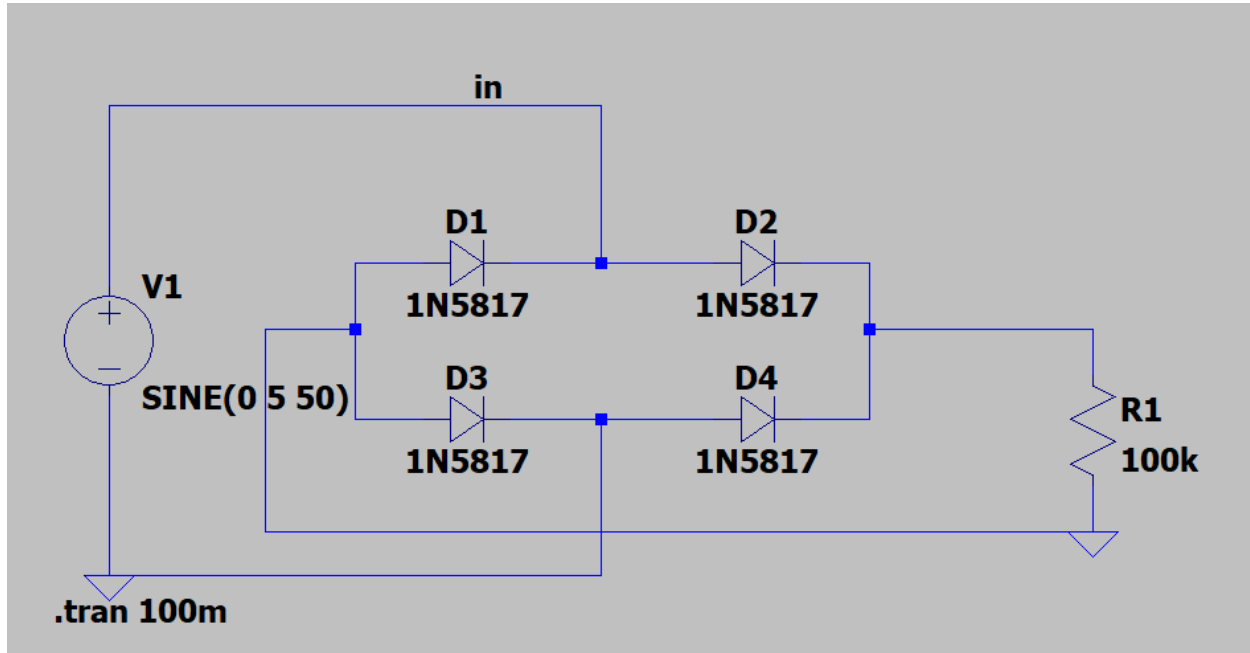
$$V_{drop \text{ per diode}} = \frac{0.8}{2} = 0.4V$$

Hence, with lower resistance, more current passes through the circuit, and there is a larger forward voltage drop through each diode.

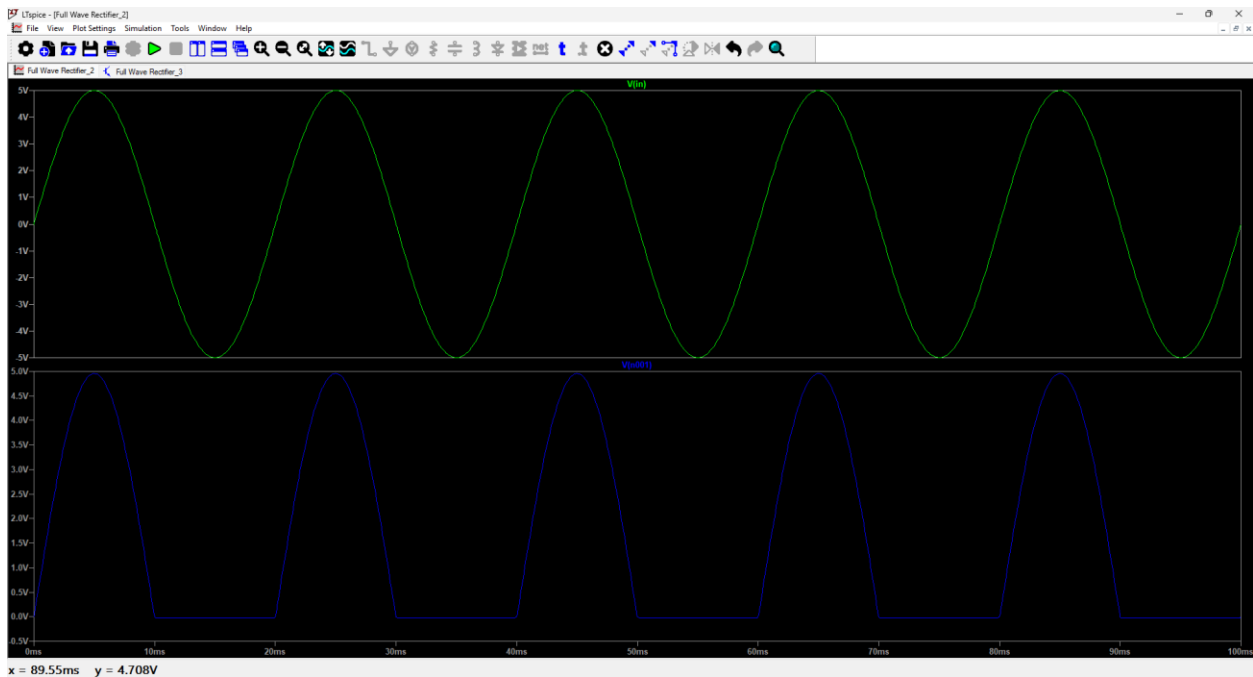
Hence, load current increases $\Rightarrow$ diode drop voltage increases $\Rightarrow$ output peak decreases.

Step 2: Full-wave rectifier using Schottky diodes.

Now we are asked to build the same circuit with the same configuration however with Schottky diodes.

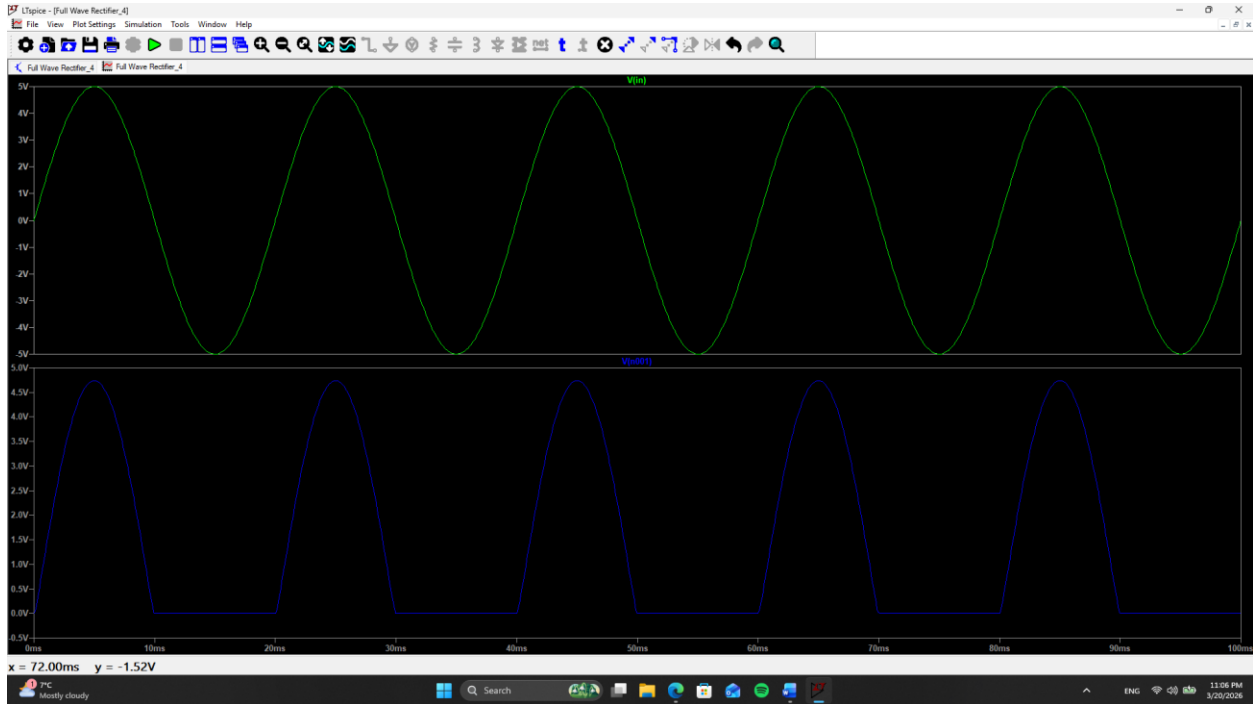


Just like in the previous step, we are asked to start with the 100kΩ resistor.



As we can see the output wave is full wave rectified, however in comparison to the previous step, the peak voltage is not affected. This is explainable by the difference of standard silicon diodes and Schottky diodes. The Schottky diodes have higher efficiency and lower power loss, they need less forward voltage to turn on. Hence the drop is much smaller, and the output peak voltage is higher when Schottky diodes are used.

Lastly, we are asked to replace the $100\text{k}\Omega$ resistors with 100Ω resistors and use Schottky diodes again.



As we can see the full wave rectified output has a peak voltage of approximately 4.75V.

$$V_{drop} = 5V - 4.75V = 0.25V$$

$$V_{drop \text{ per diode}} = \frac{0.25}{2} = 0.125V \text{ per diode}$$

This value of voltage drop across the diode is much smaller than the one we got for the regular silicone diodes. This confirms that Schottky diodes have much lower forward voltage drop.

Hence the output voltage is higher, and less voltage is lost after the rectification, making this circuit more efficient.

The only reason we saw difference in 100Ω circuit and no peak voltage difference in $100\text{k}\Omega$ circuit is due to the change in current. When the load is large enough it causes the current to be very small which results in a very small forward voltage drop across the diodes. Therefore, the output voltage almost appears unchanged for the circuit with higher load resistance.